

Bose–Einstein Condensation in Biological Systems

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Biological systems contain dipolar elements which are capable of electric oscillations at certain specific frequencies. Longitudinal electric modes in a frequency range of 10^{11} – 10^{12} s⁻¹ are typical. Fröhlich has suggested that if energy is supplied above a critical rate to the branch or branches of electric modes, Bose–Einstein condensation into the lowest energy state occurs. This phenomenon possibly represents a channel for various life processes such as cell division and protein synthesis.

Fröhlich's model is analyzed utilizing microscopic techniques, and it is subsequently concluded that Bose condensation in biosystems may occur under certain conditions.

1. Introduction

The possible participation of the vibrational modes of macromolecules in biological processes has been discussed repeatedly (Volkenshtein, 1964; Khurgin, 1967). It is well known that biological systems contain components capable of electric dipolar oscillations. For example, cell membranes possess a thickness on the order of 10^{-6} cm which maintains a strong dipolar layer. Local vibrations within the membrane could cause positively and negatively charged segments to vibrate against each other yielding an electric dipole oscillation. Similarly, biological macromolecules, e.g. proteins, DNA, etc. have marked dipolar characteristics due primarily to the presence of hydrogen bonds. The interaction between these giant dipoles through long range Coulomb forces establishes a branch or several branches of longitudinal electric modes in a frequency range of 10^{11} – 10^{12} s⁻¹.

Recently, H. Fröhlich (Fröhlich, 1968*a,b*, 1970, 1972, 1975), has suggested that if energy is supplied locally to one component (or removed from there), then, due to the long range Coulomb interaction, this energy change will be shared by all the components comprising the branch of electric modes. If the rate of energy supply is sufficiently large, the redistribution of the energy between the various degrees of freedom will lead to its concentration in the

lowest energy mode. This phenomenon is known in physics as Bose-Einstein condensation. This coherent oscillation of electric modes might serve not only as a method for efficient energy storage, but also as a channel for specific bioprocesses.

The biological implications of Bose-Einstein Condensation are numerous. With the onset of condensation, a strong coherent electromagnetic oscillation is established in the system, and the energy channeled into the lowest energy mode could possibly disrupt the cell membrane. Thus, this effect might be one of the energetic mechanisms crucial to the production of cell division. A Soviet investigator (Devyatkov, 1974) has reported enhanced cell division in *Rhodotorula* and *Candida* cultures after irradiation by electromagnetic waves in the frequency range appropriate to the Fröhlich model. Similarly, the energy stored by biological Bose-Condensation could be directed to other life sustaining activities such as protein synthesis.

Besides the possible important participation of Bose-Einstein Condensation in the processes of cell division and protein synthesis, a variety of other biological effects could perhaps be attributed to this phenomenon. Recently, several investigators (Kiselev & Zalyuboskaya, 1974; Sevastyanova & Vilenskaya, 1974; Smolyanskaya & Vilenskaya, 1974) have reported interesting results concerning the microwave irradiation of various biological objects in the frequency range 10^{11} – 10^{12} s⁻¹. Assuredly, the specific effects represent very complex biological processes. However, they exhibit two pronounced features: (i) they depend very strongly on frequency exhibiting typical resonance behavior, (ii) there exists a critical power supply threshold below which no effect is observed and above which there is a sharp increase in the biological effects which is essentially independent of the power supply. These features strongly support Fröhlich's Bose condensation theory as applied to biological systems. The complexity and diversity of the experimental results may reflect the possibility that different biological effects occur depending upon which dipolar elements are participating in the Bose condensation. For example, the vibrations of one type of macromolecule within the biological system would most likely produce different effects than a different type of macromolecule or the vibrations of the cell membrane itself.

Fröhlich arrived at his conclusions on the basis of rate equations. The general forms of the rate equations were dictated by requiring a Bose-Einstein distribution for thermal equilibrium when there is no power supplied. Beyond this, their forms were somewhat arbitrary. In this paper, the interaction Hamiltonian is utilized. It is subsequently shown that Fröhlich's ansätze in the rate equations are correct, and that there is a Bose-Einstein Condensation.

2. Theory

Similar to Fröhlich, it is presumed that the biological system consists of three parts: (i) oscillating units of giant dipoles occurring approximately periodically along the length of the macromolecules, (ii) the rest of the biosystem constituting a heat bath, (iii) an external energy source which couples to the oscillatory units. The interaction between the oscillatory units will produce a narrow band of frequencies, ω_i ($\omega_0 \leq \omega_i \leq \omega_{\max}$), corresponding to the normal modes of the electromagnetic oscillations.

To each mode, ω_i , a creation operator, a_i^+ , and a destruction operator, a_i , are assigned. The normal modes of oscillation will then interact with the rest of the biological system (the heat bath) which is represented by a set of independent excitation frequencies, Ω_k , associated with creation and destruction operators, b_k^+ and b_k respectively. Furthermore, the external energy source, associated with creation and destruction operators P_p^+ and P_p respectively and frequency, $\bar{\Omega}_p$, feeds into the electromagnetic oscillating units and provides impetus for initiation of the biological effects. The Hamiltonian of the biological system can be written as:

$$\begin{aligned}
 H = & \sum_i \hbar \omega_i a_i^+ a_i + \sum_k \hbar \Omega_k b_k^+ b_k \\
 & + \sum_p \hbar \bar{\Omega}_p P_p^+ P_p \\
 & + \sum_{i,k} \lambda b_k^+ a_i + \lambda^* b_k a_i^+ \\
 & + \sum_{i,j,k} (\chi a_j^+ a_i b_k^+ + \chi^* a_j a_i^+ b_k) \\
 & + \sum_{p,i} (\xi P_p a_i^+ + \xi^* P_p^+ a_i)
 \end{aligned} \tag{1}$$

where λ , χ , and ξ are coupling constants for the one quantum process, two quanta process, and the energy supply to the oscillating units respectively. The electromagnetic oscillations are bosons. Yet the excitations within the heat bath can be either bosons, e.g. phonons, or fermions, e.g. electrons.

The operators a_i^+ , a_i , b_k^+ , b_k , P_p^+ and P_p satisfy the commutation relations:

$$\begin{aligned}
 a_i a_j^+ - a_j^+ a_i &= \delta_{ij} \\
 a_i^+ a_j^+ - a_j^+ a_i^+ &= a_i a_j - a_j a_i = 0 \\
 b_k b_l^+ \pm b_l^+ b_k &= \delta_{kl}
 \end{aligned} \tag{2}$$

(- sign for Bosons and + sign for fermions)

$$\begin{aligned}
 b_k b_l \pm b_l b_k &= b_k^+ b_l^+ \pm b_l^+ b_k^+ = 0 \\
 P_p P_{p'}^+ - P_{p'}^+ P_p &= \delta_{pp'}
 \end{aligned}$$

The operators corresponding to different excitations commute with one another.

The rate of change of the number of quanta in the i th mode is given by:

$$\dot{n}_i = \frac{1}{i\hbar} [n_i, H] \quad (3)$$

with $n_i = a_i^\dagger a_i$.

Using the commutation relations in equation (2), equation (3) becomes:

$$\begin{aligned} \dot{n}_i = & -\frac{i}{\hbar} \sum_{k} (\lambda^* b_k a_i^\dagger - \lambda b_k^\dagger a_i) \\ & -\frac{i}{\hbar} \sum_{j,k} \{ \chi(a_i^\dagger a_j b_k^\dagger - a_j^\dagger a_i b_k^\dagger) + \chi^*(a_i^\dagger a_j b_k - a_j^\dagger a_i b_k) \} \\ & -\frac{i}{\hbar} \sum_{p,i} (\xi P_p a_i^\dagger - \xi^* P_p^\dagger a_i) \end{aligned} \quad (4)$$

The expectation value of $\dot{n}_i$ is given by:

$$\langle \dot{n}_i \rangle = \langle \psi(t) | \dot{n}_i | \psi(t) \rangle \quad (5)$$

where the angular brackets refer to thermal averaging in the grand ensemble describing the whole system, and $|\psi(t)\rangle$ is the state vector of the whole system at time t . Since equation (5) cannot be evaluated exactly, perturbation theory is utilized to solve it to the lowest order in coupling strength. To this end, it is presupposed that the interactions between the electromagnetic oscillations of the system with the heat bath and energy source are turned on adiabatically from the past. Then, if $|\psi_0\rangle$ is the exact eigenstate of the non-interaction system with the Hamiltonian

$$\begin{aligned} H_0 = & \sum_i \hbar \omega_i a_i^\dagger a_i + \sum_k \hbar \Omega_k b_k^\dagger b_k \\ & + \sum_p \hbar \bar{\Omega}_p P_p^\dagger P_p, \end{aligned}$$

it is then possible to write the state vector at time t , to the lowest order in coupling constants as:

$$\begin{aligned} |\psi(t)\rangle = e^{-iH_0 t/\hbar} & \left[1 - \sum_{i,k} \frac{\lambda b_k^\dagger a_i e^{i(\Omega_k - \omega_i)t}}{(\omega_i - \Omega_k) + i\delta} \right. \\ & - \sum_{i,k} \frac{\lambda^* b_k a_i^\dagger e^{-i(\Omega_k - \omega_i)t}}{(\omega_i - \Omega_k) - i\delta} \\ & - \sum_{i,j,k} \frac{\chi a_j^\dagger a_i b_k^\dagger e^{i(\Omega_k + \omega_j - \omega_i)t}}{(\Omega_k + \omega_j - \omega_i) + i\delta} \\ & - \sum_{i,j,k} \frac{\chi^* a_j a_i^\dagger b_k e^{-i(\Omega_k + \omega_j - \omega_i)t}}{(\Omega_k + \omega_j - \omega_i) - i\delta} \\ & - \sum_{p,i} \frac{\xi P_p a_i^\dagger e^{i(\omega_i - \bar{\Omega}_p)t}}{(\omega_i - \bar{\Omega}_p) - i\delta} \\ & \left. - \sum_{p,i} \frac{\xi^* P_p^\dagger a_i e^{-i(\omega_i - \bar{\Omega}_p)t}}{(\omega_i - \bar{\Omega}_p) + i\delta} \right] |\psi_0\rangle \end{aligned} \quad (6)$$

Where δ is an infinitesimally small positive value, changing the summations over Ω_k and $\bar{\Omega}_p$ to integrations and using appropriate contours yields:

$$|\psi(t)\rangle = e^{-iH_0 t/\hbar} \left[1 - 2i\pi \sum_i (\lambda b_i^+ a_i - \lambda^* b_i a_i^+) - 2i\pi \sum_{i,j} (\chi a_j^+ a_i b_{i-j}^+ + \chi^* a_j a_i^+ b_{j-i}) - 2i\pi \sum_i (\xi P_i a_i^+ + \xi^* P_i^+ a_i) \right] |\psi_0\rangle \quad (7)$$

Substituting equation (7) into equation (5) and using the commutation relations, one obtains:

$$\begin{aligned} \langle \dot{n}_i \rangle = & S - \phi(T, \omega_i) [\langle n_i \rangle e^{\hbar\omega_i/kT} - (1 + \langle n_i \rangle)] \\ & - \sum_j \Lambda(T, \omega_i, \omega_j) [\langle n_i \rangle (1 + \langle n_j \rangle) e^{[\hbar(\omega_i - \omega_j)]/kT} \\ & - \langle n_j \rangle (1 + \langle n_i \rangle)] \quad (8) \end{aligned}$$

where $\langle n_i \rangle$ is the thermal average of the number of i th quanta and $S = 4\pi/\hbar |\xi|^2 \sum_i \langle P_i^+ P_i \rangle_0$ is the number of quanta supplied to the i th mode from the external source. $\phi(T, \omega)$ and $\Lambda(T, \omega_i, \omega_j)$ are given by:

$$\begin{aligned} \phi(T, \omega_i) &= 4\pi/\hbar |\lambda|^2 e^{-(\hbar\omega_i/kT)} [1 \pm N(\hbar\omega_i)] \\ \Lambda(T, \omega_i, \omega_j) &= 2\pi/\hbar |\chi|^2 e^{-[\hbar(\omega_i - \omega_j)]/kT} \begin{cases} [N(\hbar\omega_j - \hbar\omega_i)], & \text{for } \omega_j > \omega_i \\ [1 \pm N(\hbar\omega_i - \hbar\omega_j)], & \text{for } \omega_i > \omega_j \end{cases} \quad (9) \end{aligned}$$

where $N(\hbar\omega)$ is the number of excitations with energy $\hbar\omega$ in the heat-bath and the plus or minus sign corresponds to bosons or fermions respectively. Note that the 2nd and 3rd terms in equation (8) possess exactly the same forms as Fröhlich's ansätze for loss rates of i th quanta in one and two quanta processes respectively.

The condition for stationarity requires for each oscillator $\langle \dot{n}_i \rangle = 0$. From this, it is found from equation (8) that:

$$\langle n_i \rangle = \left[1 + \frac{S}{\phi(T, \omega_i) + \sum_j \Lambda(T, \omega_i, \omega_j) \langle n_j \rangle} \right] e^{[(\hbar\omega_i - \mu)/kT]^{-1}} - 1, \quad (10)$$

and

$$e^{-(\mu/kT)} = 1 - \frac{\sum_j \Lambda(T, \omega_i, \omega_j) [\langle n_j \rangle - (1 + \langle n_j \rangle) e^{-(\hbar\omega_j/kT)}]}{\phi + \sum_j \Lambda \langle n_j \rangle} \leq 1. \quad (11)$$

The inequality of equation (11) together with the requirement that $\langle n_i \rangle \geq 0$ dictates that:

$$\hbar\omega_0 \geq \mu \geq 0.$$

3. Conclusion

Equations (10) and (11) are precisely the same equations obtained by Fröhlich. Clearly, from equation (10) one sees that Bose Condensation into the lowest energy mode occurs when μ is approximately equal to $\hbar\omega_0$. As demonstrated by Fröhlich, this takes place when the energy supply becomes sufficiently large and exceeds a certain specific rate which depends on the coupling strength of the heat bath. If there is no energy supplied, $s = 0$, $\mu = 0$, and equation (10) becomes the thermal equilibrium distribution as required. Also it is interesting to take note that if there are no two quanta processes, i.e. $\chi = 0$, then $\mu = 0$, and no Bose Condensation occurs even if energy is supplied externally.

By approaching this theoretical problem using microscopic techniques, a better understanding of the possible existence of Bose-Einstein Condensation in biological systems has been obtained. This model, therefore, is a useful tool for gaining insight into intricate biological structures via straightforward physical concepts. However, the explicit understanding of the complex biochemical aspects of this problem and how they relate to specific life processes is yet to be developed.

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